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Costumer Segmentation and Churn-Risk Analytics

A data-driven framework for customer segmentation, behavioral risk classification, explainability and business recommendations

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Executive Summary

Logistic S.p.A. requires a structured analytical framework to transform customer, shipment, sector and geographic data into actionable intelligence. The project addresses this problem by developing an end-to-end customer analytics pipeline that cleans and enriches data, engineers behavioral variables, identifies customer segments, detects anomalies, assigns customer health and risk indicators, and prepares a supervised churn-risk classification layer. The analysis combines K-Means clustering as the main segmentation model and DBSCAN as a complementary anomaly-detection layer. Customer profiling translates numerical clusters into interpretable business groups, while the Customer Risk Score, Customer Health Index and customer status labels support monitoring and prioritization. The supervised learning phase uses Logistic Regression, Random Forest and XGBoost to classify a rule-derived churn-risk target. Random Forest is selected as the preferred operational model in the executed analysis because it provides the strongest balance between recall, F1-score and overall accuracy. SHAP explainability confirms that risk classification is mainly driven by behavioral activity and trend variables, especially recent ratio, volume trend, activity drop, trend percentage and recent volume. The final framework supports targeted retention, anomaly investigation, account prioritization and future dashboard-based monitoring.

Keywords: customer segmentation; churn-risk classification; K-Means; DBSCAN; SHAP explainability.

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1. Introduction and Business Challenge

The project addresses the need for Logistic S.p.A. to obtain a clearer understanding of its customer base. The available data contain customer characteristics, monthly shipment or activity volumes, company size indicators, sector information and geographic fields. However, these raw variables do not directly indicate which customers are strategically important, which are declining, which are stable, and which require managerial attention.

The analytical objective is to build a coherent customer intelligence framework able to segment customers, detect atypical behavior, profile segments in business terms, identify risk conditions and support future monitoring. The work does not claim to observe real future churn, because an externally validated future churn label is not available. Instead, it defines a transparent rule-derived churn-risk target based on historical inactivity, decline and anomaly patterns.

- **Business problem:** customer heterogeneity makes it difficult to prioritize retention, growth and monitoring actions.
- **Analytical objective:** transform raw operational and firmographic data into interpretable customer segments and risk indicators.
- **Managerial objective:** provide Logistic S.p.A. with a framework for differentiated customer actions, early-warning monitoring and explainable decision support.

2. Data Understanding, EDA and Cleaning

The raw dataset contains 20,436 rows and 42 columns, including customer sector information, legal nature, revenue, employees, province and 36 monthly activity columns. The EDA phase was used to verify whether the data was reliable and suitable for customer segmentation and churn-risk analysis. The analysis started by checking the dataset structure, variable types, missing values, duplicated records and inconsistencies in categorical and numerical fields. Missing or problematic values were handled to reduce noise before modeling. Sector information was enriched through the **ATECO-to-NACE mapping**, while geographic information was prepared to support later regional profiling. The most important outcome of the EDA was the confirmation that the monthly activity columns could be transformed into meaningful behavioral indicators. These variables made it possible to describe each customer in terms of value, recent activity, historical activity, trend, volatility and deterioration. Therefore, EDA provided the cleaned and enriched analytical base for the following feature engineering, clustering and risk-scoring phases.

Tab. I – EDA and cleaning operations

EDA component	Role in the project
Dataset structure	20,436 rows and 42 columns were inspected to understand the available information.
Monthly activity variables	36 monthly columns were identified and used to reconstruct customer behavior over time.
Missing values and inconsistencies	Missing, empty and inconsistent fields were cleaned or converted before modeling.
Numerical conversion	Revenue, employees and monthly activity values were converted into numeric format.
ATECO to NACE logic	Italian ATECO sector codes were mapped to the corresponding NACE categories using the official ISTAT correspondence table. This standardized the sector information and made customer profiling by industry more consistent and interpretable.
Geographic enrichment	Province information was used to support regional and geographical profiling.

3. Feature Engineering and Data Pre-Processing

Feature engineering started from the original dataset columns. Revenue and employees were used to describe company size, while the 36 monthly activity columns were used to reconstruct customer behavior over time. From these original variables, the final **11 modeling features** reported in **Tab. II** were created to summarize the main analytical dimensions needed for clustering: company scale, customer value, recent engagement, average activity, trend, volatility and deterioration. As shown in **Tab. II**, the feature matrix combines firmographic and behavioral information. `log_revenue` and `log_employees` describe customer size; `log_total_volume`, `log_recent_volume` and `log_avg_monthly_volume` capture activity intensity; `volume_trend` and `trend_pct` measure growth or decline; `log_volatility` and `cv_volume` describe activity instability; while `recent_ratio` and `activity_drop_ratio` capture recency and deterioration signals relevant for churn-risk monitoring. Before clustering, the preprocessing operations summarized in **Tab. III** were applied. Column names were harmonized, monthly activity columns were detected and ordered chronologically, and revenue, employees and monthly values were converted into numeric format. Missing and infinite values were handled to avoid modeling errors. *Log transformation* was applied to skewed size and volume variables to reduce the effect of extremely large customers. *Capping* was applied to trend and ratio variables at the 1st and 99th percentiles to limit the distortion caused by extreme values while preserving their information. Finally, all 11 features were *standardized*, because K-Means and DBSCAN are distance-based models and require variables to be on comparable scales. The correlation matrix in **Fig. 1** was used as a diagnostic check of the final feature set. Some correlations are expected, especially among volume-related variables, but the matrix confirms that the selected features still cover distinct business dimensions: size, value, recency, trend, volatility and deterioration.

Tab. II – Final feature modeling matrix used for K-Means and DBSCAN

Feature	Analytical dimension	Meaning and rationale
`log_revenue`	**Company size / value**	Log-transformed revenue. It represents business value while reducing skewness and the impact of extremely large companies.
`log_employees`	**Company size / scale**	Log-transformed number of employees. It approximates operational scale and reduces the effect of extreme employee counts.
`log_total_volume`	**Historical customer value**	Log-transformed total shipment/activity volume across the observation period.
`log_recent_volume`	**Recent engagement**	Log-transformed recent activity volume, used to capture current commercial activity.
`log_avg_monthly_volume`	**Average intensity**	Log-transformed average monthly activity, summarizing the regular activity level.
`volume_trend`	**Absolute trend**	Difference between recent average activity and past average activity.
`trend_pct`	**Relative trend**	Percentage change between recent and past activity, useful to compare customers with different scales.
`log_volatility`	**Activity instability**	Log-transformed standard deviation of monthly activity.
`cv_volume`	**Relative volatility**	Coefficient of variation, measuring instability relative to average activity.
`recent_ratio`	**Recency concentration**	Share of total customer activity concentrated in the recent period (i.e., last 6 months)
`activity_drop_ratio`	**Deterioration signal**	Intensity of recent decline compared with expected activity based on past behavior.

Tab. III – Data pre-processing operations before clustering

Step	Purpose
Column harmonization	Italian column names were mapped into modeling names, e.g. Fatturato -> Revenue, Dipendenti —> Employees, Provincia —> Province.
Monthly column detection	Monthly activity variables M-35 to M-0 were detected and sorted chronologically.
Numerical conversion	Revenue, employees and monthly volumes were converted to numeric variables.
Missing/infinite handling	Infinite values were replaced and missing numerical values were imputed with the median.
Capping	Extreme values in trend and ratio variables were clipped at the 1st and 99th percentiles.
Log transformation	Skewed size and volume variables were transformed with log1p.
Standardization	The final 11 features were standardized before K-Means and DBSCAN to make distance calculations comparable.

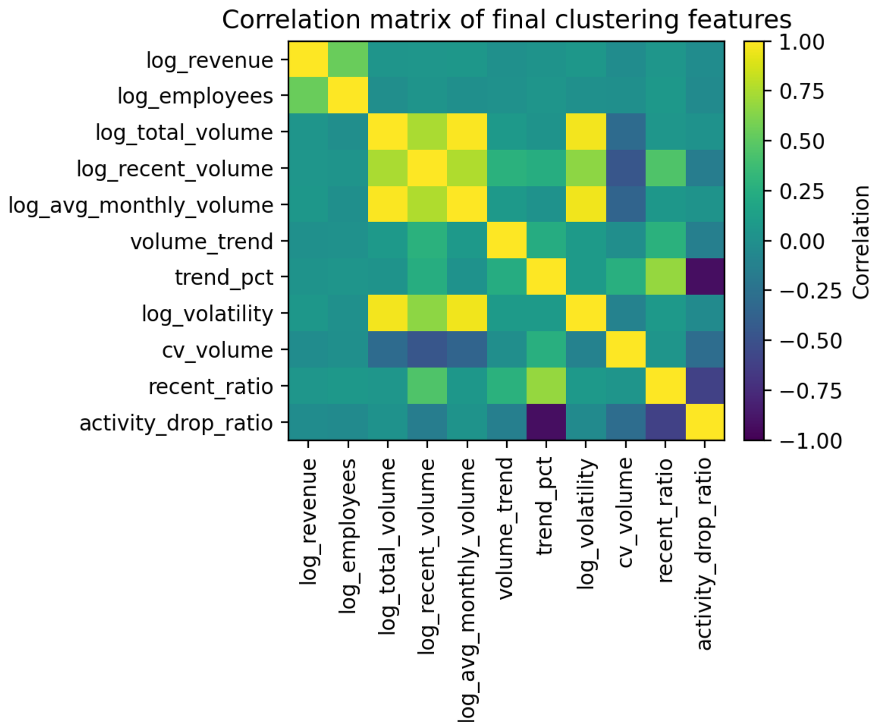

Fig. 1 – Correlation matrix of the final clustering features.

After cleaning and preprocessing, the final modeling dataset contains **16,791 customer records**. This dataset is used consistently for K-Means, DBSCAN, customer profiling, risk scoring and customer status labeling.

4. Unsupervised Machine Learning for Customer Segmentation

Unsupervised machine learning was applied to identify hidden customer structures without using a predefined target variable. This is appropriate because the project does not start from an observed future churn label; the first objective is to understand how customers differ based on historical activity, company size and behavioral indicators. Two complementary algorithms were used: **K-Means** and **DBSCAN**. As summarized in **Tab. IV**, K-Means is a centroid-based algorithm that assigns every customer to one of K clusters, making it suitable for building the main customer segmentation layer. DBSCAN, instead, is density-based and can label sparse observations as noise or anomalies. Therefore, K-Means is used to create broad and interpretable customer groups, while DBSCAN is used to detect atypical customers that do not fit the main behavioral structure. The two methods are not alternatives in this project; they provide complementary views of the customer base.

Tab. IV – Conceptual comparison between K-Means and DBSCAN

Aspect	K-Means	DBSCAN
Main logic	Centroid-based clustering.	Density-based clustering.
Primary role in the project	Main segmentation layer.	Anomaly and density-structure layer.
Assignment behavior	Assigns every customer to one of K clusters.	Can label sparse observations as noise/anomalies.
Key parameters	K, the number of clusters.	eps and min_samples.
Business interpretation	Creates broad customer groups for differentiated strategy.	Flags customers that behave atypically and deserve review.

4.1 K-Means Clustering

K-Means was applied to the standardized 11-feature matrix developed in the previous section, **Tab. II**. Since the algorithm is distance-based, standardization was necessary to ensure that all variables contributed comparably to the clustering process. The analysis compared K from 2 to 8, but the three main candidate solutions are K = 3, K = 4 and K = 5. The objective was not only to maximize statistical metrics, but to select a segmentation that was stable, interpretable and useful for business action. The comparison metrics in **Tab. V** and **Fig. 2** show that each solution has advantages and limitations:

- K = 3 obtains the highest silhouette score and the lowest Davies-Bouldin index, which indicates a compact and simple structure. However, this solution is too coarse for the business objective: it risks merging behaviorally distinct customers into broad groups, reducing the usefulness of the segmentation for differentiated actions. K = 3 is statistically cleaner but less informative from a managerial perspective.
- K = 5 provides more granularity and further reduces inertia, as expected when the number of clusters increases. However, the additional cluster introduces more fragmentation without a proportional gain in interpretability. As shown in **Fig. 3**, the K = 5 solution creates smaller and less stable groups, making the segmentation more difficult to translate into clear business guidelines. For this reason, K = 5 is not wrong, but it is less appropriate for a practical customer management framework.
- The K = 4 solution provides the best compromise between statistical quality and business readability. Although its silhouette score is lower than K = 3, the cluster-size comparison in **Fig. 3**, the relative business-profile heat maps in **Fig. 4**, and the PCA visualizations in **Fig. 5** show that K = 4 creates more meaningful behavioral separation. It avoids the excessive simplification of K = 3 while also avoiding the fragmentation of K = 5.

The final K = 4 segmentation, reported in **Tab. VI**, identifies four interpretable customer groups: **Churn-risk / declining**, **Recent-growth / opportunity**, **Stable / lower-intensity customers**, and **Strategic / high-value active**.

Key outcome: K-Means provides the main segmentation layer of the project. The K = 4 solution is considered the most appropriate because it balances model diagnostics, cluster size, PCA separation and business interpretability. This segmentation becomes the basis for customer profiling, risk scoring and differentiated business recommendations.

Tab. V – K-Means comparison metrics for K = 3, K = 4 and K = 5

K	Inertia	Silhouette	Davies-Bouldin	Calinski-Harabasz	Smallest cluster	Largest cluster	Largest cluster share %
3	105,837.945	0.301	1.086	6,254.623	287	9,049	53.892
4	93,142.178	0.249	1.220	5,500.544	287	6,398	38.104
5	81,610.591	0.270	1.111	5,301.014	275	6,373	37.955

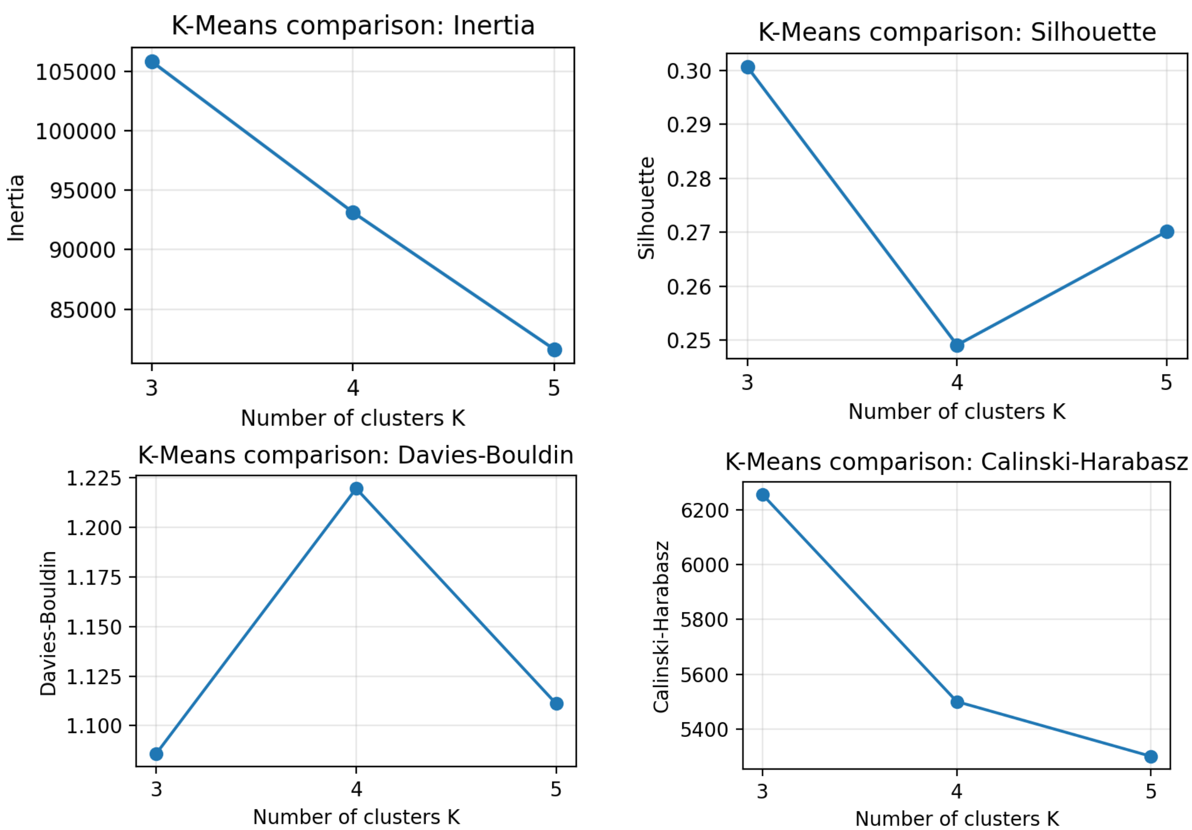

Fig. 2 – K-Means model-selection metrics used to compare alternative values of K.

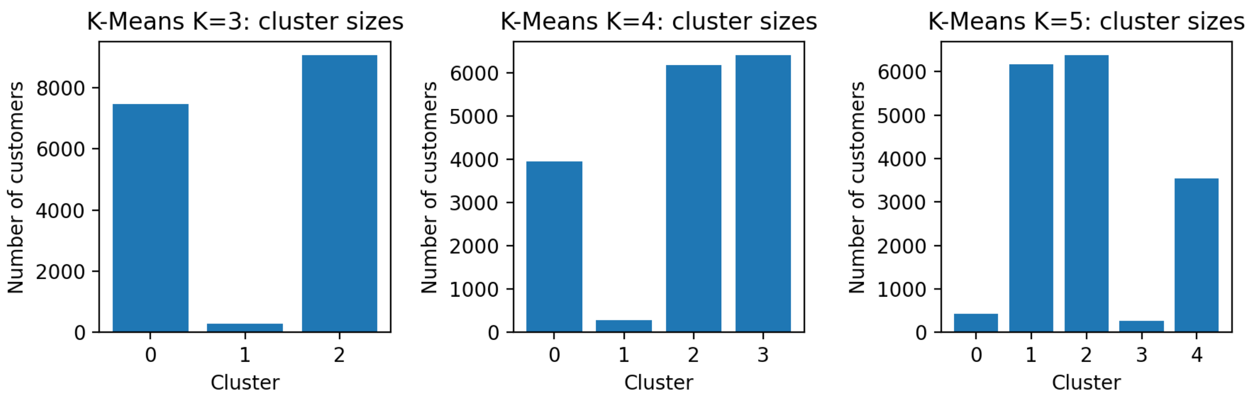

Fig. 3 – Cluster-size comparison for K = 3, K = 4 and K = 5.

Fig. 4 – Relative business-profile heat maps for alternative K-Means solutions.

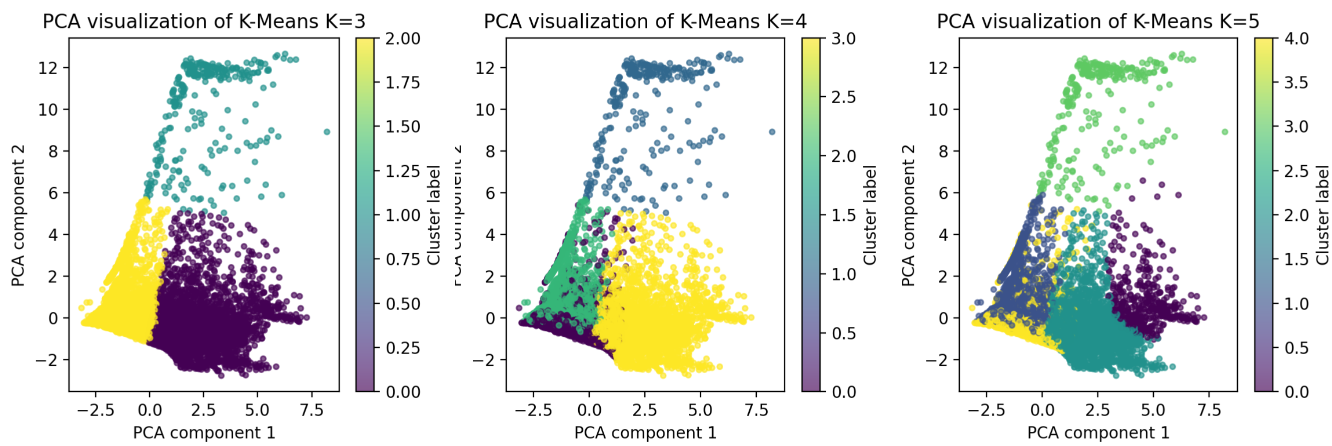

Fig. 5 – PCA visualizations of the K-Means cluster structure for K = 3, K = 4 and K = 5.

Tab. VI – Final K-Means K = 4 segment distribution

Cluster	K-Means Segment	Customers (#)	Share (%)
0	**Churn-risk / declining**	3,941	23.471
1	**Recent-growth / opportunity**	287	1.709
2	**Stable / lower-intensity customers**	6,165	36.716
3	**Strategic / high-value active**	6,398	38.104

4.2 DBSCAN Anomaly Detection

DBSCAN was applied as a complementary unsupervised method to K-Means. While K-Means assigns every customer to one of the predefined clusters, DBSCAN is density-based: it groups customers located in dense areas of the feature space and labels isolated observations as *noise*. DBSCAN is therefore not used to replace the K-Means segmentation, but to add an *anomaly-detection layer*. This is useful because some customers may belong to a K-Means segment but still show atypical behavior compared with the majority of the customer base.

The two key DBSCAN parameters are **eps** and **min_samples**. The **eps** parameter defines the maximum distance within which neighboring customers are considered part of the same dense region. The **min_samples** parameter defines the minimum number of nearby observations required to form a dense area. Since the clustering matrix contains 11 modeling features, **min_samples** was set to 22, following the practical rule of using approximately twice the number of features, making the density criterion sufficiently robust in a multi-dimensional feature space.

The selection of **eps** was guided by the k-distance diagnostics reported in **Tab. VII** and **Fig. 6**. The k-distance curve shows the distance between each customer and its 22nd nearest neighbor, sorted from lowest to highest. The curve remains relatively smooth for most customers and then rises sharply in the right-hand tail. This sharp increase indicates observations that are far from dense regions and are therefore potential anomalies. The 90th percentile threshold was selected as a stricter value because the objective was not to force DBSCAN to create broad clusters, but to preserve its ability to detect atypical customer behavior. Higher thresholds, such as the 95th or 97th percentiles, would be more permissive and would classify more customers as part of dense regions, reducing the anomaly-detection value of the model.

Tab. VII – DBSCAN k-distance percentile diagnostics

min_samples	q80	q85	q90	q92	q95	q97
9	0.604	0.691	0.815	0.895	1.082	1.315
11	0.642	0.731	0.863	0.949	1.149	1.419
22	0.782	0.890	1.064	1.171	1.414	1.738
33	0.871	0.995	1.203	1.321	1.593	1.952

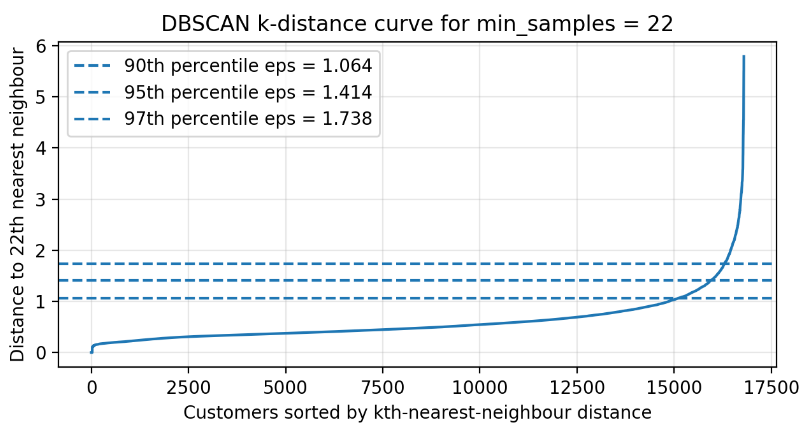

Fig. 6 – DBSCAN k-distance curve used to guide eps selection.

Tab. VIII – Final DBSCAN label distribution

Label	DBSCAN Group	Customers (#)	Share (%)
-1	**Noise / atypical customer**	828	4.931
0	**Minor dense subgroup 1**	123	0.733
1	**Dominant dense customer opulation**	15,776	93.955
2	**Minor dense subgroup 2**	34	0.202
3	**Minor dense subgroup 3**	30	0.179

The final DBSCAN distribution is shown in **Tab. VIII**. Most customers are assigned to the main dense population, while 828 customers, corresponding to about 4.9% of the dataset, are labeled as *noise or atypical customers*. The small additional dense subgroups indicate the presence of local behavioral structures, but the main analytical value of DBSCAN in this project is the identification of the noise group. The PCA representation in **Fig. 7** confirms that DBSCAN separates the broad dense customer population from observations located in sparse or peripheral areas of the feature space.

Key outcome: DBSCAN adds a second layer of interpretation to the K-Means segmentation. Customers labelled as noise should not automatically be interpreted as negative or lost customers; rather, they are customers with unusual behavioral patterns. For this reason, the DBSCAN noise flag is used later in customer profiling, risk scoring and status labeling as an additional warning signal for customers requiring further business review.

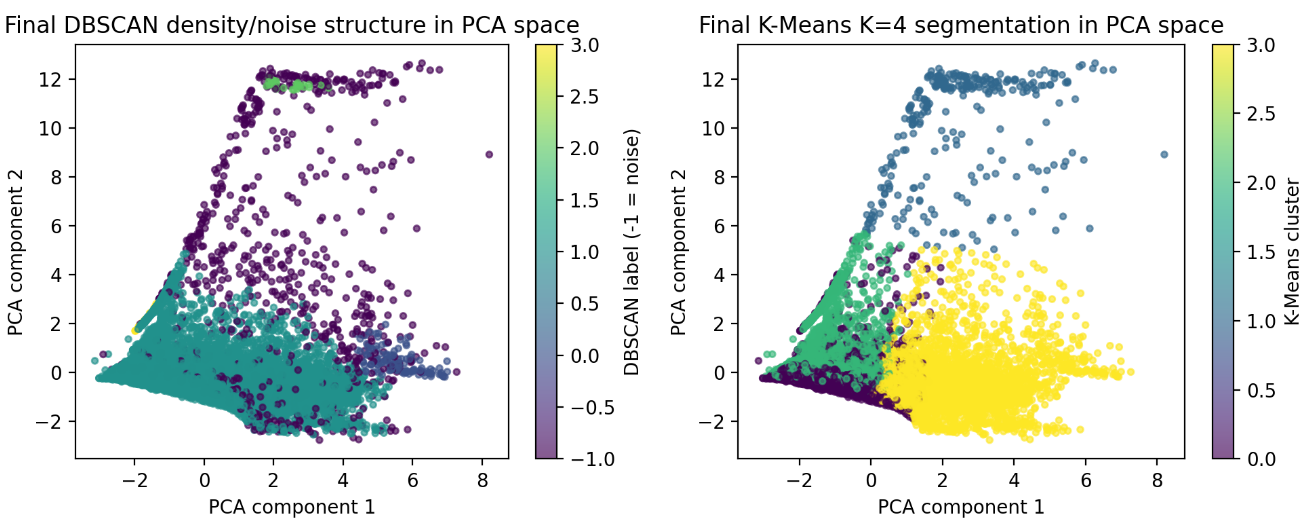

Fig. 7 – Final DBSCAN density/noise (left) and K-Means K=4 (right) structure represented in PCA space.

4.3 Summary

The unsupervised layer combines two complementary perspectives. K-Means provides the main segmentation structure by assigning each customer to one of four business-readable groups. As shown in **Fig. 8**, the largest segments are **Strategic / high-value active** and **Stable / lower-intensity customers**, followed by the **Churn-risk / declining** segment. The **Recent-growth / opportunity** segment is much smaller, but strategically relevant because it captures customers with emerging activity patterns. DBSCAN adds a second layer by identifying customers whose behavior is atypical compared with dense customer populations. The right-hand chart in **Fig. 8** shows that the DBSCAN noise share is not distributed uniformly across K-Means segments. The most relevant insight is that the **Recent-growth / opportunity** segment has the highest share of atypical customers. This suggests that many customers in this segment behave differently from the rest of the customer base: they may represent genuinely emerging opportunities, irregular new activity, or customers requiring closer validation before being interpreted as stable growth cases. The other segments show much lower DBSCAN anomaly shares. This means that **Strategic / high-value active**, **Stable / lower-intensity**, and **Churn-risk / declining** customers are mostly located within denser and more regular behavioral regions. Therefore, their K-Means profiles are more structurally stable, while the recent-growth segment requires more careful interpretation.

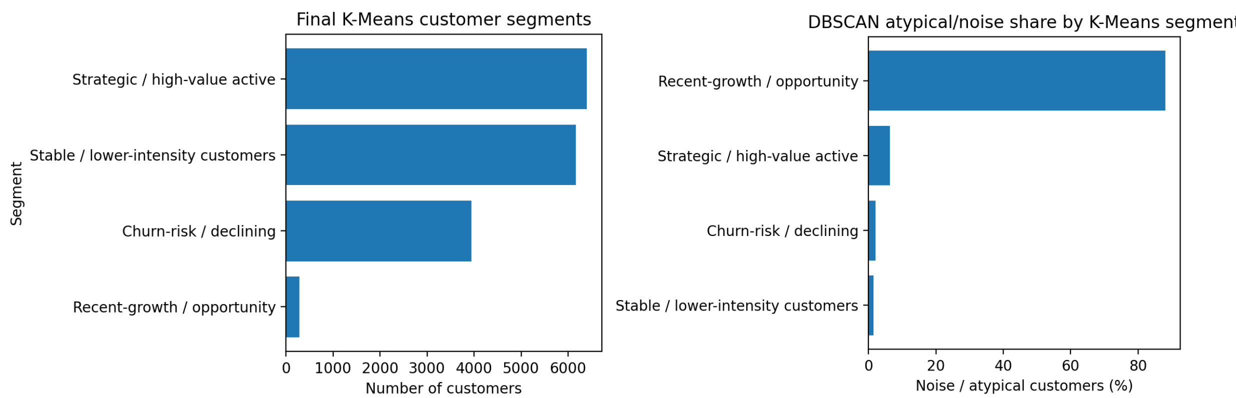

Fig. 8 – Final K-Means customer segment sizes (left) and DBSCAN atypical/noise customer share by K-Means segment (right).

Key outcome: K-Means defines the main customer segmentation map, while DBSCAN highlights atypical behavior within and across those segments. Together, they provide a stronger framework than either model alone: K-Means supports strategic segmentation, while DBSCAN adds anomaly monitoring and helps identify customers that should be reviewed separately during profiling and risk assessment.

5. Customer Profiling and Risk-Based Labeling

5.1 Customer profiling

Customer profiling translates the K-Means cluster labels into business meaning. A cluster number is not actionable by itself; it becomes useful only when it is described through variables that managers can interpret, such as customer volume, recent activity, trend, revenue, employees, volatility and DBSCAN anomaly share. For this reason, profiling is the bridge between the unsupervised learning output and the business interpretation of customer segments. As shown in **Tab. IX**, the four K-Means segments present clearly different behavioral profiles:

- **Churn-risk / declining** segment shows weak recent activity and a negative trend, suggesting customers whose relationship with the company is deteriorating;
- **Recent-growth / opportunity** segment is small, but it has very high recent activity concentration and the highest DBSCAN noise share, indicating customers with unusual but potentially valuable emerging behavior.
- **Stable / lower-intensity** segment includes a large share of customers with limited but relatively regular activity.
- **Strategic / high-value active** segment contains the largest and most commercially relevant active customers, characterized by high total volume and positive activity dynamics.

This profiling step is important because it converts the segmentation into differentiated business logic. Each segment requires a different managerial interpretation: declining customers should be monitored for retention, recent-growth customers should be validated and developed, stable low-intensity customers can be managed efficiently, and strategic high-value customers should be protected and prioritized.

Tab. IX – Segment profiling summary by K-Means customer segment.

Metric	Churn-risk / declining	Recent-growth / opportunity	Stable / lower-intensity customers	Strategic / high-value active
K-Means cluster	0	1	2	3
Customers (#)	3,941	287	6,165	6,398
Customer share (%)	23.471	1.709	36.716	38.104
Avg. Revenue	43,661.624	5,608,010.143	6,552,347.474	4,773,153.543
Avg. Employees	2.687	25.369	32.322	29.299
Avg. Total Volume	1,077.213	6,007.606	262.406	41,009.293
Avg. Recent Volume	71.238	5,667.017	13.432	7,002.191
Avg. Trend (%)	-0.238	242.533	-0.085	0.317
Avg. Recent Ratio	0.086	0.974	0.113	0.166
Avg. Activity Drop Ratio	0.068	-1,208.398	-0.736	-0.334
DBSCAN Noise Share (%)	2.055	88.153	1.460	6.314

Notably, negative values of `activity_drop_ratio` indicate that recent activity is higher than past activity; therefore, strongly negative values should be interpreted as growth signals rather than deterioration.

Key outcome: Customer profiling transforms the K-Means clusters into actionable customer groups, providing the basis for the subsequent risk scoring, Customer Health Index and status labeling framework.

5.2 Risk Scoring and Customer Health Index

Segmentation explains how customers differ, but it does not directly indicate which customers require immediate attention. For this reason, a **Customer Risk Score** and a **Customer Health Index** were introduced after profiling to transform segment interpretation into an operational monitoring framework. The Customer Risk Score summarizes the intensity of negative behavioral signals, while the Customer Health Index expresses the same information in the opposite and more business-friendly direction: higher health indicates a stronger and less risky customer relationship. The two indicators adapt established churn-risk scoring practices to Logistic S.p.A.'s data through a custom rule-based framework based on activity, trend, volatility, DBSCAN anomaly status and K-Means segment context.

In this project, the **Customer Risk Score** is a synthetic rule-based indicator. It does not represent financial default risk or an observed probability of future churn. Instead, it summarizes behavioral warning signals derived from customer activity history and unsupervised-learning outputs. As shown in **Tab. X**, the score combines six main components: low recent activity, activity drop, negative trend, high volatility, DBSCAN anomaly status and K-Means segment context. These components capture different forms of customer deterioration, such as reduced engagement, declining volume, unstable activity or atypical behavior.

Each component is first converted into a comparable 0–100 risk scale and then combined through the following weighted framework:

$$\begin{aligned}\text{Customer Risk Score} = {} & 0.25 \cdot \text{Recent Activity Risk} + 0.20 \cdot \text{Activity Drop Risk} \\ & + 0.20 \cdot \text{Trend Risk} + 0.15 \cdot \text{Volatility Risk} \\ & + 0.10 \cdot \text{DBSCAN Anomaly Risk} + 0.10 \cdot \text{Segment Context Risk}\end{aligned}$$

Recent activity risk increases when recent volume is low. Activity drop risk increases when the customer shows a stronger decline compared with past behavior. Trend risk increases when the activity trend is weak or negative. Volatility risk increases when customer activity is unstable.

DBSCAN anomaly risk flags customers classified as noise or atypical, while segment context risk adjusts the score according to the behavioral risk profile of the K-Means segment.

The **Customer Health Index** is then calculated as:

$$\text{Customer Health Index} = 100 - \text{Customer Risk Score}$$

This means that a high risk score corresponds to a low health index, while a high health index indicates a more stable and less risky customer relationship. The segment-level results reported in **Tab. XI, Fig. 9** and **Fig. 10** show that risk is not distributed uniformly across K-Means segments. The **Stable / lower-intensity customers** segment has the highest average Risk Score, equal to **59.51**, the lowest average Customer Health Index, equal to **40.49**, and the highest high-risk share, equal to **33.40%**. This suggests that customers with lower activity intensity should not automatically be interpreted as healthy, because weak or limited activity may still indicate deterioration or disengagement.

The **Churn-risk / declining** segment also shows elevated risk, with an average Risk Score of **55.78**, an average Health Index of **44.22**, and a high-risk share of **30.14%**. This is coherent with the segment interpretation, since these customers are characterized by weaker recent activity and declining behavioral signals. By contrast, the **Recent-growth / opportunity** segment shows a much healthier profile, with a high-risk share of **0.00%** and an average Health Index of **66.11**. The **Strategic / high-value active** segment has the highest average Health Index, equal to **69.97**, and a low high-risk share of **5.47%**, confirming its overall positive profile. However, this segment also contains **275 strategic at-risk customers**, meaning that valuable customers may still require targeted monitoring when early warning signals appear.

Tab. X – Components of the Customer Risk Score.

Risk component	**Variables**	**Business meaning**
Low recent activity	`recent_volume, recent_ratio`	Signals weak current engagement.
Activity drop	`activity_drop_ratio`	Captures deterioration relative to historical activity.
Negative trend	`volume_trend` `trend_pct`	Identifies customers moving toward lower activity.
High volatility	`cv_volume` `log_volatility`	Captures unstable and less predictable activity.
DBSCAN anomaly	`is_dbscan_noise / dbscan_label`	Flags atypical patterns that do not fit dense customer behavior.
Segment context	`kmeans_segment`	Adjusts interpretation according to the customer group.

Tab. XI – Risk summary by K-Means segment

Metric	Stable / lower-intensity customers	Churn-risk / declining	Recent-growth / opportunity	Strategic / high-value active
K-Means cluster	2	0	1	3
Customers (#)	6,165	3,941	287	6,398
Customer share (%)	36.72	23.47	1.71	38.10
Avg. Health Index	40.49	44.22	66.11	69.97
Avg. Risk Score	59.51	55.78	33.89	30.03
High-Risk Share (%)	33.40	30.14	0.00	5.47
Strategic at-risk customers (#)	2	78	0	275
Avg. Value-at-Risk Score	16.08	22.74	16.71	23.51

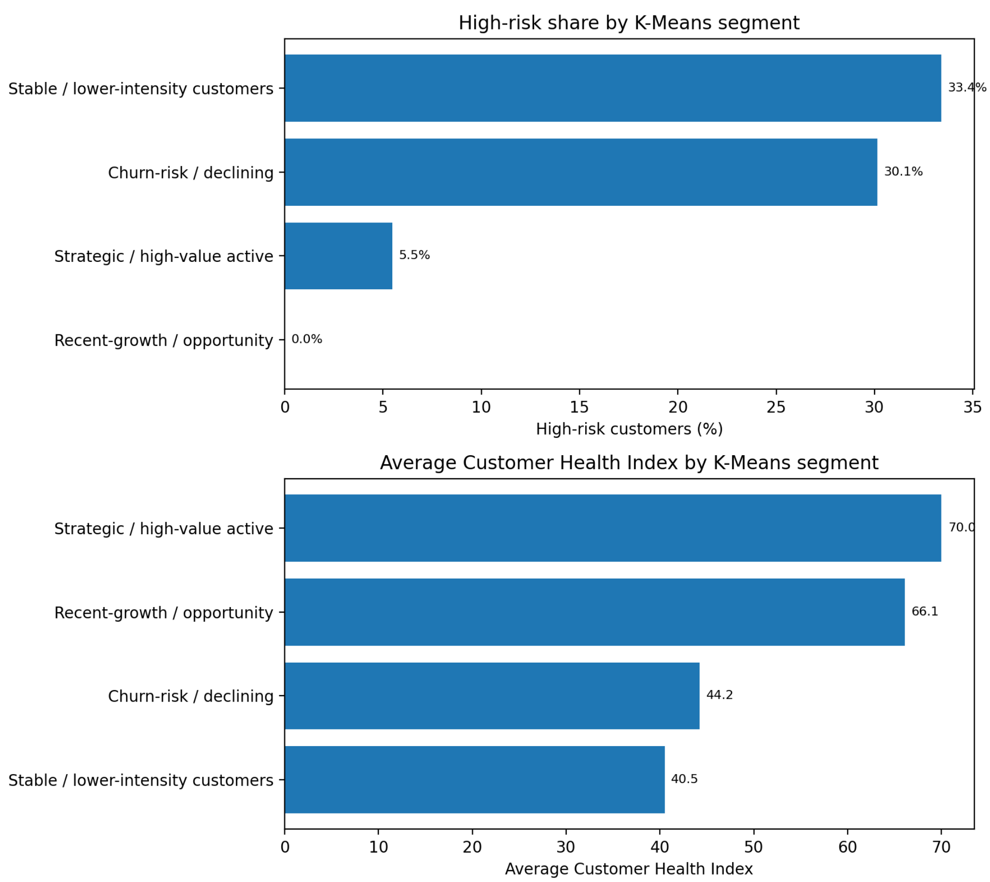

Fig. 9 – High-risk share by K-Means segment (top), average Customer Health Index by K-Means segment (bottom).

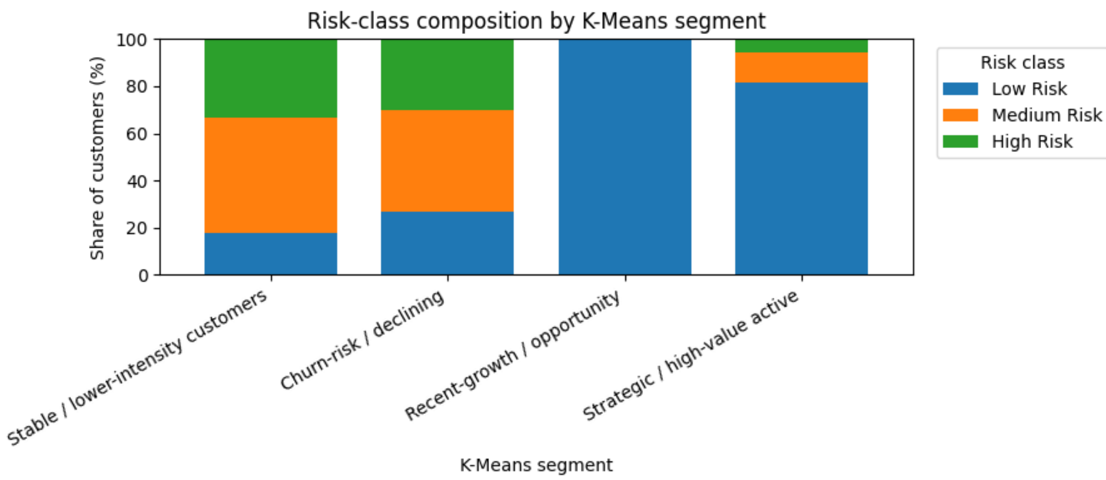

Fig. 10 – Risk-class composition by K-Means segment.

Fig. 10 confirms the same interpretation from a risk-class perspective. The Stable / lower-intensity and Churn-risk / declining segments contain the largest proportions of medium-risk and high-risk customers. The Recent-growth / opportunity segment is classified as low risk, while the Strategic / high-value active segment is mostly low risk but still includes a smaller share of medium- and high-risk customers.

Key outcome: The risk and health framework makes the segmentation actionable. It shows which customer groups require retention, monitoring, reactivation or protection actions, and it prepares the basis for the following customer status labeling.

5.3 Customer status labeling and rule-derived target

Customer status labels were created to translate numerical indicators into business-readable categories. Instead of relying only on raw risk scores, the labeling framework assigns each customer to an interpretable status that can support account-management decisions. The labels are based on customer activity, decline, growth, volatility, anomaly status and risk intensity. Since the project does not have an externally validated future churn label, these categories are not treated as observed churn outcomes. Instead, they represent a **rule-derived churn-risk proxy** based on historical inactivity, deterioration and anomaly patterns. As shown in **Tab. XII**, the status labels distinguish between high-risk proxy conditions, monitoring conditions, healthy customers, growth opportunities and cases requiring manual review.

Tab. XII – Customer status labels and business interpretation.

Status label	Definition	Interpretation
Lost_or_Inactive	No meaningful recent activity despite historical activity.	High-risk proxy
Declining	Customer shows relevant decline or deterioration signals.	High-risk proxy
Anomalous_High_Risk	Atypical DBSCAN behavior combined with high-risk evidence.	High-risk proxy
Volatile_Monitor	Customer is active but unstable and should be monitored.	Monitoring
Stable_Healthy	Customer shows stable, recent activity and lower risk.	Healthy
Active_Growth	Customer shows active and growing behavior.	Opportunity
Unclassified_Check	Customer does not clearly fit the previous business rules.	Manual review

The high-risk proxy labels are:

- **Lost_or_Inactive**, for customers with no meaningful recent activity despite historical activity;
- **Declining**, for customers showing relevant deterioration signals;
- **Anomalous_High_Risk**, for customers combining atypical DBSCAN behavior with high-risk evidence.

The monitoring label is **Volatile_Monitor**, for active but unstable customers that should be observed over time.

The lower-risk or opportunity-oriented labels are:

- **Stable_Healthy**, for customers with stable recent activity and lower risk;
- **Active_Growth**, for customers showing active and growing behavior.

Finally, **Unclassified_Check** is retained for customers that do not clearly match the predefined rules and may require further business validation. The resulting distribution is reported in **Tab. XIII** and visualized in **Fig. 11**.

The largest group is `Lost_or_Inactive`, with **7,048 customers**, corresponding to **41.97%** of the dataset. It is followed by `Stable_Healthy`, with **3,500 customers**, or **20.84%**, and `Active_Growth`, with **3,284 customers**, or **19.56%**.

Tab. XIII – Customer status distribution.

Customer status label	Customers	Share (%)
Lost_or_Inactive	7,048	41.97
Stable_Healthy	3,500	20.84
Active_Growth	3,284	19.56
Unclassified_Check	1,493	8.89
Volatile_Monitor	1,346	8.02
Anomalous_High_Risk	104	0.62
Declining	16	0.10

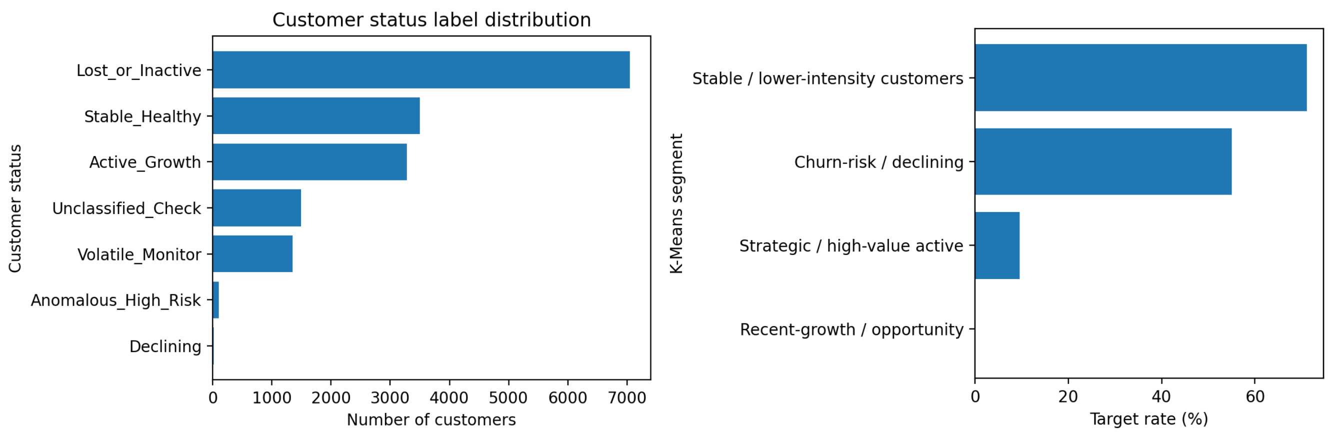

Fig. 11 – Customer status label distribution (left) and rule-derived churn-risk target rate by K-Means segment (right).

The `Unclassified_Check` group includes **1,493 customers** or **8.89%**, while `Volatile_Monitor` includes **1,346 customers** or **8.02%**. The smaller high-risk categories are `Anomalous_High_Risk`, with **104 customers** or **0.62%**, and **Declining**, with **16 customers** or **0.10%**.Overall, the three high-risk proxy categories — `Lost_or_Inactive`, `Declining` and `Anomalous_High_Risk` — include **7,168 customers**, corresponding to **42.69%** of the final dataset. These customers form the rule-derived high-risk target used for the following supervised-learning phase.

Key outcome: Customer status labeling converts the analytical outputs into operational business categories. It allows Logistic S.p.A. to distinguish customers to reactivate, monitor, protect, develop or manually review. In addition, because no real future churn label is available, the high-risk proxy labels provide a transparent rule-derived target for the next supervised classification step.